

# AI531C — Week 6 Pre-Class Activity (PCM-6: Three Point Pairs)

Prepares: Lecture 14 (Registration & homography, Tue 8 Sep) · Lectures 15–16 (Edges; Thresholding & watershed), Wed 9 Sep · Tutorial T6 (Tue 8 Sep).

## 1 · Topic

Aligning two images of the same scene: how many correspondences buy which transformation, and what RANSAC rescues when clicks go wrong.

## 2 · Why you are preparing this

Lecture 14 builds registration live from clicked points. If the parameter-counting arithmetic (how many pairs for affine? for homography?) is already yours, class becomes about failure — wrong clicks, outliers, and the RANSAC vote.

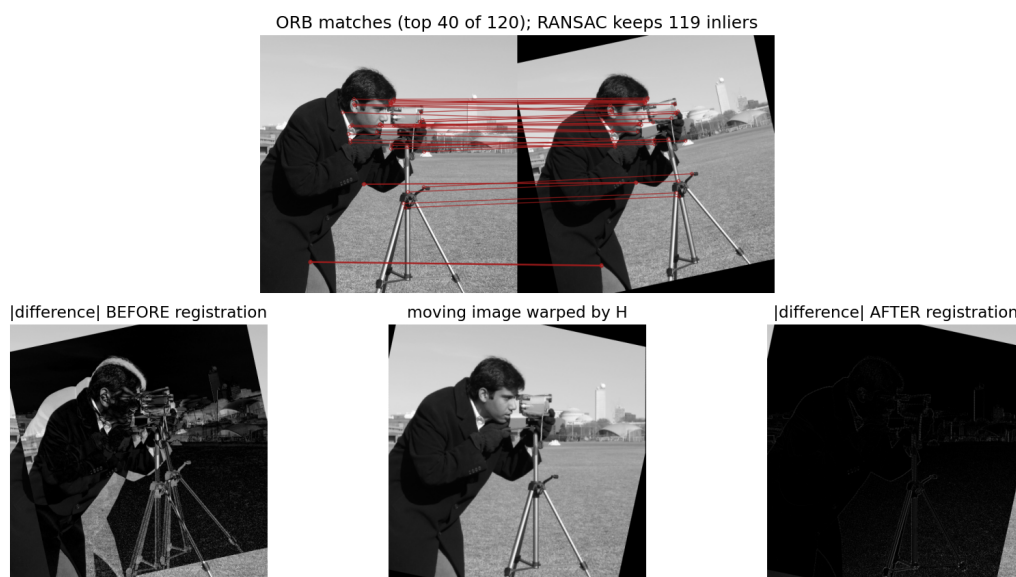
## 3 · Read (12–15 min)

From the Comprehensive Self-Contained Lecture Notes:

- Lecture 14 chapter, the sections on affine vs projective transforms and on correspondence (about 3 pages);
- Lecture 15 chapter, ONLY the gradient definition page (edges start Wednesday).

*Optional depth: Szeliski Ch. 8 (alignment), figures only.*

## 4 · Interpret one figure (4 min)



*Source: course figure, Lecture 14 (registration before/after panel).*

In two sentences: what evidence in the difference image tells you the alignment worked, and what residue would a small rotation error leave?

## 5 · One small calculation (5 min)

An affine transform has 6 unknowns; each point pair gives 2 equations. Compute the minimum pairs for affine, and repeat for a homography (8 unknowns). Why do practitioners use MANY more than the minimum?

## 6 · Predict (4 min)

Predict: if ONE of your clicked pairs is wrong by 30 pixels, does least-squares fitting stay near the truth or follow the error? What property of squared error decides this?

## 7 · Inspect this fragment (3 min)

```
H, mask = cv2.findHomography(pts1, pts2, cv2.RANSAC, 5.0)
print(mask.sum(), 'of', len(pts1), 'pairs kept')
```

Question: what is RANSAC doing with the pairs it did NOT keep, and why is 'kept count' worth printing every time?

## 8 · Bring one question to class

One question about alignment — or two photos of yours that you would like to overlay (bring them; T6 can try).

**Submit: items 4-6 — half a page maximum, before Tue 8 Sep, 9:00 AM.**

Expected time: 30-35 minutes. Counts toward the weekly preparation portfolio (best 10 of PCM-2...PCM-14, completion-based micro-credit per the lesson plan). The short per-lecture primers in the course folder are optional warm-ups; only THIS card is submitted.